

# **Study About the Binary Option Pricing Formula With Hidden Markov Model**

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**Abstract:** Due to the influence of two different factors, macro environment and micro interference, we have derived a more objective stock option asset pricing recurrence formula, by simulating the macro market environment with a hidden Markov model of three zone system transfer, and representing the micro price characteristics in different markets with binary tree random variables. Further, compared of the option pricing results with the classic **binary** tree models by the examples of SSE 50 ETF. It is found although the set of hidden Markov regime switching model and the calculation of asset price are more complicated, but the results are more in line with the objective reality. The option pricing formula are worth of reference by the issues of financial asset pricing and risk management. We also give some application features, and precautions regarding the application of conclusions.

**Key Words:** Regime-Switching; Hidden Markov Model; Binary Tree Model; Option Pricing

## 1. Introduction

Traditional stock price model such as Binomial tree model and Black-Scholes model assume that the discrete form of stock price stochastic process is approximated by stochastic process of continuous events, this assumption has been accepted by researchers in the short term and has become a standard in the financial community. While, empirical studies have shown that such models are often ineffective in capturing extreme price motion problems(Hardy, 2001). Therefore, the scene makers maintain that this method may not be able to deal with the relevant optimization, combinatorial analysis and other related problems effectively, especially the downside risk issue which is often considered in decision making. There are many improved models for such situation, among those models, the autoregressive conditional heteroskedasticity (ARCH) model and the generalized autoregressive conditional heteroskedasticity (GARCH) model have been widely accepted, and these models can be regarded as a special form of regime-switching model. In short, regime-switching Model tries to capture the discrete changes of the financial variables by allowing the parameters of the model to take different values in different periods with the changes of the market. Therefore, this kind of model breaks the thinking mode of the traditional model, and is

more in line with the objective reality. The Hidden Markov model is another form of regime switching model which has attracted widespread attention in recent years. The main characteristic of this model is Bivariate property, using implicit state variables which is unobservable but can affect the parameters of the variable of financial assets that can be observed. Markov chain (or Markov process) represents the different stages of the implied economic status or economic cycle, different values showed different market states, so that, Regime-switching model can perform the structure of economic conditions. Compared with the heteroskedasticity model, this potential Markov process can be reflected in any variable parameters of modern asset price model, whether it is a parameter representing the trend item or a parameter representing the variance. So the Hidden Markov model is more objective to describe the market capacity.

The history of regime-switching model can be traced back to the work of Quandt (1958), Goldfeld and Quandt (1973), they applied a class of regime-switching model to nonlinear economic data. The idea of this model is also used in some other earlier nonlinear time series analysis (Tong (1983)). Hamilton (1989) took the lead in applying the regime-switching model to the study of finance and econometrics. Empirical studies showed that this kind of model can better fit economic and financial time series and explain the important structural phenomena in these series. Thus, there are many different application forms of regime-switching model in the financial domain, such as the research of Pliska (1997) and Elliott et al. (2001) on the short interest rate models, the study on combinatorial analysis by Elliott and Hinz (2002), the research on option pricing by Naik (1993), Guo (2001) and Elliott et al. (2005), the study on variation estimation by Elliott et al. (1998) and so on. Salah G U et al.(2018)decomposed oil price changes into supply, demand, and risk driven shocks and used Markov regime switching model to analyze the nonlinear effects of oil price shocks. Pan X et al.(2019)took Bangladesh as an example, by using annual data from 1972 to 2017, the study was conducted using a Markov switching vector error correction model, and the results reveal a two-regime switching nature of the effect of financial development on energy intensity. Jia Wang et al. (2021) based on the

theoretical framework of expected utility with uncertain probabilities, a Markov regime switching model is proposed, which divides the stock market into bear and bull states, and tests whether there are significant differences in their ambiguity degrees. The results show that the degree of ambiguity of the Chinese mainland stock market is significantly higher than that of the Hong Kong stock market, and there are significant differences between bear and bull states for the two markets. Ping et al. (2023) tracked and studied the motion of stocks and other asset price process under the conditional Black-Scholes model, they also derived and analyzed an option pricing formula for the Black-Scholes asset model under the macro environment of the transfer of the two zones system.

Hidden Markov model is a form of regime switching model has been widespread concerned in recent years. The main character of this model is Bivariate property, use implicit state variables which cannot be observed to affect the parameters of the variable of financial assets that can be observed. Markov chain(or Markov process) represents the different stages of the implied economic status or economic cycle, different values showed different market statuses, so that, Regime-switching model can performance the structure of economic conditions. Compare with heteroskedasticity model, this potential Markov process can be reflected in any variable parameters of modern asset price model, whether it is a parameter representing the trend item or a parameter representing the variance. So the Hidden Markov model is more objective to describe the market capacity.

In recent years, more and more people pay attention to the application of the regime-switching model to study the option pricing problem, especially the problem when the economic conditions are objective to structural changes. For example, stock linked embedded option securities and participating life insurance products etc, the selection of long-term stock is very important. However, due to the uncertain information caused by the regime switching, the market under the model is incomplete, so in this condition, there is more than an equivalent martingale measure for pricing problems. Therefore, the standard Black-Scholes-Moton option pricing formula cannot be directly applied. In the incomplete market, the choice of equivalent martingale

measure is very important, and many methods have been developed. Among these methods, there are different select methods of improved measures for regime-switching Model has been used. For example, Guo(2001) applied a set of "virtual" assets, i.e., the state transition contract method is for option pricing of Regime-switching market with continuous time. Elliott et al. (2005) has proposed using the Esscher transform method in option pricing of regime-switching Model with continuous time. Siu (2002) applied further improved transfer method in option pricing of regime-switching Model with continuous time. Ishijima and Kihara (2005) studied option pricing problem of regime-switching Model driven by Hidden Markov model under discrete time, they applied the equivalent martingale measure of local risk neutral value from Duan (1995). Chuin et al.(2010) applied Esscher transform model and a class of extended Girsanov principle respectively selected the discrete hidden Markov regime-switching under Gauss Model for the option pricing model. Ma J, Wang H(2019) studied the convergence rates of moving mesh methods for a system of moving boundary partial integro-differential equations (PIDEs) which arised in the Asian option pricing under the state-dependent regime-switching jump - diffusion models. The study introduced a moving finite difference method (FDM) to solve the one-dimensional moving boundary problem and the convergence rates were proved. Xiaoyuan Z, Tianqi Z (2022) considered the pricing of barrier options under a Markov regime switching diffusion model. The proposed model incorporated structural changes in economic conditions and business and investment environments into the diffusion process and by implementing extensive numerical experiments, they present that the derived solution was highly effective and tight. Liu X, Zhang Z(2023) considered stochastic default barriers driven by a regime-switching model and geometric Brownian motion, improved upon the assumption of fixed default barriers, then proposed an enhanced model for pricing vulnerable options and they derived an explicit pricing formula for European vulnerable options. Ping et al.(2023) studied the Black-Scholes Asset Price under the Regime-Switching Framework, and got the consult is compared to a simple random asset price model, the regional transfer asset price model with convertible macro environment values has better fitting ability point by point.

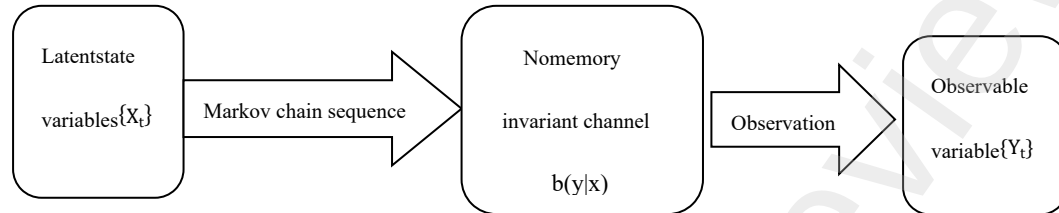
As we know, binary tree model is a simple description to the state of the market, because the model is relatively simple and easy to calculate, and under the assumption of discrete time interval is sufficiently small, the option pricing formulas of binary tree model and continuous Black-Scholes Model are consistent. Therefore, binary tree model is widely recognized and applied by researchers. In this paper, we mainly consider the stock option pricing problem under the discrete binary tree model driven by the hidden Markov Model which represents the market state, and we assume there 3 different market states. Which fit the actual market rules more really. The model also can avoid the problems caused by the choice of measures in the incomplete market under binary tree model. We firstly give the hidden Markov Model and the transition law between market states. Then, we give the basic hypothesis about the operating rules of asset prices in three types of macro market state transitions. After clarifying the operating rules of asset prices in the transition between market states, we further provide the operating formula of market prices and derive the discounted price recursive formula for future options with dual uncertainties of macro market and micro impact. In order to examine the conclusion of this article, in empirical research, we applied the Shanghai Stock Exchange 500 Index to conduct a comparative analysis of option pricing with the classical binary difference model. The results showed that the pricing method obtained in this article is superior to the classical binary tree model.

The full text is divided into five parts: first, introduction, put forward and combine the problem. Second, introduction about the hidden Markov model. Third, give the hypothesis in this paper. Fourth, the derivation of the option pricing formula with regime-switching under the condition that the variable and unobservable market is represented by the Markov model. Fifth, a comparison with the simple binary tree option pricing formula under the condition of single market. At last, summarize the conclusion, application features, and precautions.

## **2. Hidden Markov Model**

In simple words, the hidden Markov model is a bivariate model. In these two random variables, one is a potential state variable or can be regarded as dependent variable, generally which meets characteristics of homogeneous Markov model, but the

value of the variable can not be observed. Another variable is observable, theoretically, the value of the variable is mainly affected by the potential state variables which cannot be observed. Through Figure 1 , we can have an intuitive understanding of Hidden Markov model.



**Figure 1 Presentation for the relation between the variables of Hidden Markov process**

Specifically, let  $\{X_1, X_2, \dots\}$  be a Markov Chain with discrete time, which values in the finite dimensional state space  $X$ .  $M$  is the number of the states in this space. Without losing generality, we assume that  $X = \{1, 2, \dots, M\}$ , let  $X$  present the possible values of  $X_t$ . Let  $\pi = \{\pi_i\}$  is a  $1 \times M$  dimensional vector which serves as the initial probability distribution. Markov chain is usually assumed to be homogeneous in time, let  $\alpha_{ij} = P(X_t = j | X_{t-1} = i)$  is transition probability, and  $P = \{\alpha_{ij}\}$  is a  $M \times M$  dimensional transition probability matrix. Consider that a discrete time channel process has input variables  $\{X_1, X_2, \dots\}$  and output variables  $\{Y_1, Y_2, \dots\}$  for each time  $t$ .  $Y_t$  value in observable space  $Y$ . Assume that the channel has no memory and invariable, for any given  $X_t$ , let  $b(y_t | x_t)$  be the corresponding channel transfer density.

Hence, the density function of  $n$ -dimensional variable  $(y^n, x^n)$  is

$$P(y^n, x^n) = P(y_1, x_1) \prod_{t=2}^n P(y_t, x_t | x_{t-1}). \quad (1)$$

Where  $P(y_1, x_1) = \pi_{x_1} b(y_1 | x_1)$ , and  $P(y_t, x_t | x_{t-1}) = \alpha_{x_{t-1}x_t} b(y_t | x_t)$ ,  $t = 2, 3, \dots$ .

For all  $x_j \in X$ ,  $j = 1, 2, \dots, M$ , set  $\alpha_{x_0x_j} = \pi_{x_j}$ , then, the density function of  $n$ -dimensional observable variable  $Y^n$  is

$$P(y^n) = \sum_{x^n} \prod_{t=1}^n \alpha_{x_{t-1}x_t} b(y_t | x_t). \quad (2)$$

In reality research and application, generally only the samples of observable variables can be obtained, the potential state variables cannot be observed, the corresponding transition probability matrix and the transition density function of non-memory invariant channel are required according to the specific information to the corresponding hypothesis analysis, and then need to use the parameter estimation and hypothesis test method.

### 3. Basic Assumption

The traditional Binary tree model did not take market into account the state of the , only assumed the pricing of the asset (stock) price: assuming in each period, the price of the stock only has two possible changes, rise or fall, and the range of the rise or fall in each period is the same as the probability of their occurrence. However, in reality, the market is changing over time. Sometimes the market has a good trend, sometimes the market has a bad tendency, and sometimes there is no news, people maintain a wait-and-see attitude. In this case, the probability that a stock will rise or fall is different, and the range of the rising or falling may be different. In this paper, we only consider the situation of the market exists state transition and the difference between the range of the stock's rising and falling is not discussed.

The same with binary tree model, in this article, we assume the changes of the stock price are discrete, only at the beginning or the end of the time interval may the price changes may occur, there are two trends: a rise and fall, rising and falling. Let  $S$  be the price of the stock, then assume that asset price process is  $\{S_t\}$ .  $t = 0, 1, 2, \dots$ , is the point time when the stock price may be changed with uniform and equal intervals.

According to the fact there are three cases of market expansion, contraction and stabilization. We assume the market has three kinds of states which have a different impact on the stock prices: first, there is no new situation, the market maintains the early state. Second, there is news that benefits the development of the market, the stock price will rise. Third, there is news which is not conducive to the development of the market, the stock price will fall. We use  $-1, 0, 1$  to express the three conditions which are unfavorable, stable and conducive to the development of the market, at any time the market will be able to reflect only one of these three states. Let random variables  $X$  be

the state of the market.  $X_t$  is the state of the market at time  $t$ . Since the three states of the market are incompatible with each other, and at any time, there must be only one of the three, so there will be some definite transition rules between the three states, in the model, we note the possibility of the transition between the states to the transition probability. Thus, we have the following transition probability matrix

$$P = \begin{pmatrix} P_{-1-1} & P_{-10} & P_{-11} \\ P_{0-1} & P_{00} & P_{01} \\ P_{1-1} & P_{10} & P_{11} \end{pmatrix}$$

Where  $P_{ij}$  is the probability that the market changes from state  $i$  to state  $j$ ,  $i, j = -1, 0, 1$ , and in the transition probability matrix, the sum of the elements of any rows

is 1. Since the market state is based on the development of under market of three conditions set, unfavorable, stable and favorable. Then theoretically, in these three market conditions, the probability stock rising and falling will be different. Thus, if the current stock price is  $S_t$ , then the next time will be:

- (1) The probability the rising of the stock price in the next time is  $u^{-1}$ , and the probability of decline is  $1 - u^{-1}$  under the condition that the market is in the unfavorable state -1;
- (2) The probability the rising of the stock price in the next time is  $u^0$ , and the probability of decline is  $1 - u^0$  under the condition that the market is in the maintain state 0;
- (3) The probability the rising of the stock price in the next time is  $u^1$  and the probability of decline is  $1 - u^1$  under the condition that the market is in the favorable state -1.

However, no matter what the market situation is, we assume that if the stock price rises in next future time, it will be promoted to  $hS_t$  and if it falls, it will be reduced to  $kS_t$ . That is, we assume that the price of a stock rises or falls at the same rate in any market situation. In particular, if  $u^{-1} = u^0 = u^1$ , our model is reduced to the general binary tree pricing model. In other words, the simple binary tree pricing model is a special case of our model.

An important assumption for a simple binary tree model is that the market interest rate always keeps a fixed constant that does not change. In order to meet the three

market conditions and meet the basic condition of no arbitrage, the interest rate will also have three constant forms. As with the simple binary tree model, they will match the corresponding stock price and the probability of price changes in different market conditions. Respectively, note that the market interest rates are  $r^{-1}$ ,  $r^0$  and  $r^1$  in the three market states. In this paper, we use the superscripts for all variables to represent the state of the market, and subscripts to state (time).

Since the model is complicated by the market actual situation, we are concerned with the stock price situation after several equal time periods in the future, whether it will be different with the stock prices that will be obtained under the simple binary tree model, and further we care about the option pricing formula for the stock price. In order to make the problem clearer, further we make a simpler explanation of other assumptions:

(1) As with the simple binary tree model, we assume that the stock price is  $S_0$  at the beginning of the period (zero time), and assume that the market state at the beginning (zero time) is known. This paper assumes that the beginning of the market condition is stable.

(2) Assume that both the market state and the stock price change at the same time interval (Figure 2). That is, the state of the market or the price of the stock may change at discrete time points, and that between any two adjacent time points the stock price or market condition will continue to change from the previous time point until the next possible time point for the arrival of the change.

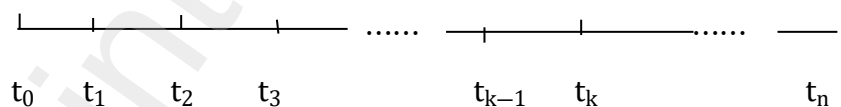


Figure 2. A diagram of the time point where the market may change

In Fig. 2, we assume that the time point  $t_0$  is the initial time,  $t_1, t_2, \dots, t_k, \dots, t_n$  are respectively the time points in the future, which are likely to change, and they are all equal time intervals, for any  $i \neq j, i, j = 1, 2, \dots, n$ ,  $t_i - t_{i-1} = t_j - t_{j-1}$ . So that all the time period in which we are concerned with is totally divided into equal  $n$  time interval. For any  $k$ -th time interval ( $1 \leq k \leq n$ ), the time point  $t_{k-1}$  is the starting point of

the time interval, and the time point  $t_k$  is the ending point of the time interval.

(3) Assuming that future market conditions are determined at the end of the previous period, the market price trend for the next period is determined at the beginning of the period and the resulting final stock price at the end of the period. That is to say, the market situation that the market may have in the  $k$ -th time interval is determined at the end of the  $k-1$ th time interval, or it can be determined at the time point  $t_{k-1}^-$ . The possible movement of the stock price in the  $k$ -th time interval and the corresponding interest rate of the market will be determined at the time point  $t_{k-1}^+$ . Based on these conditions and current market information, the stock price of the final  $k$ -th period will be obtained at the time point  $t_k$ .

Theoretically, it can be explained that the future state of the market depends on the state of the previous period and the situation of various kinds of information. At the beginning of the new period, the state of the market was formed. While the future stock price process depends on the price of the previous period and a variety of information, of which the market situation is the most important influence. The stock prices will be changed after a variety of information, so we assume that stock price process will be chosen at the beginning of each period is in line with the theoretical and actual market conditions.

According to the nature of the Markov model, the market states in any  $k$  ( $2 \leq k \leq n$ ) periods are only determined by the situation of the previous market period and determined at the end of the previous period, so that we have

$$\begin{aligned} P(X_k = j) &= \sum_{i=-1}^1 P(X_k = j | X_{k-1} = i) = \sum_{i=-1}^1 P_{i,j} \\ P(X_k = j | X_{k-1} = i) &= P_{i,j} \end{aligned} \quad (3)$$

The corresponding market in a different states will also have different interest rate values, thus

$$P(r_k = r^j | X_{k-1} = i) = P_{i,j} \quad (4)$$

Where  $i, j = -1, 0, 1$ .

#### 4. the European call option pricing when the initial market situation is smooth

##### i. The single-time European call option pricing

This paper assumes that the initial state of the market is known, which is a stable market condition. In this case, the interest rate is  $r^0$ , the initial stock price is  $S_0$ . Since the market state at the initial moment is stationary, the stock at the end of the period may rise to  $hS_0$  with probability  $u_0$  or fall to  $kS_0$  with probability  $1 - u_0$ . If the exercise price is assumed to be  $K$ , then the call option pricing formula for this single-time market state at the known initial time is the same as for the simple case. We have

$$C_0 = \frac{u^0 C_{11} + (1 - u^0) C_{12}}{1 + r^0} \quad (5)$$

Where  $C_{11} = \max(0, hS_0 - K)$ ,  $C_{12} = \max(0, kS_0 - K)$ .

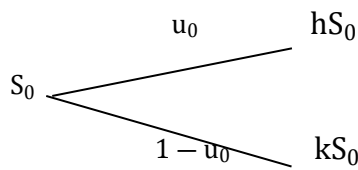


Figure 3 single-period stock price chart

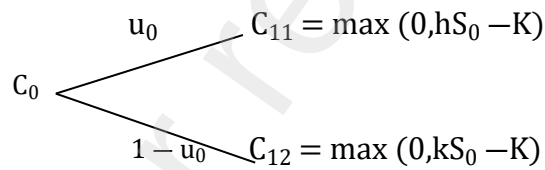


Figure 4 single-period option price chart

## ii. The European call option Pricing of second-stage

As the second-stage of the market state is not fully predict, but it can be said that at the end of the first stage and the beginning of the second stage, the market may have three possible states, and in different states of the market the probability of the stock rising or falling is no longer the same, and the corresponding market risk-free rate is not the same, which makes our options pricing more difficult.

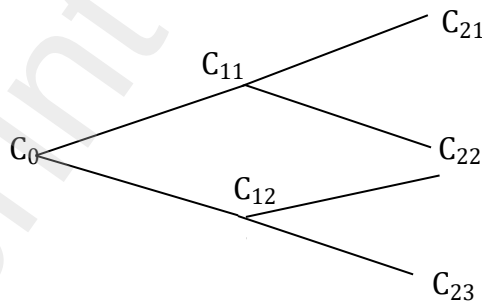


Figure 5 the chart for the simple second period options price

Figure 5 shows the possible price distribution of the option price at the end of the second period. The situation is always same as the simple binary tree option assumption.

Under the assumption of simple binary tree model, we can deduce the price of one

terminal option directly according to the following formula:

$$C_{11} = \frac{u_1 C_{21} + (1 - u_1) C_{22}}{1 + r_1}, \quad C_{12} = \frac{u_1 C_{22} + (1 - u_1) C_{23}}{1 + r_1} \quad (6)$$

Since it is assumed that no matter what the market state occurs, the rise or fall of the stock is the same, so that at the end of the second-stage, the stock may appear at the same price looks like the simple binary tree model. Thus the possible price of the option is the same as that of the simple binary tree model, which is  $C_{21} = \max(0, h^2 S_0 - K)$ ,  $C_{22} = \max(0, h k S_0 - K)$ ,  $C_{23} = \max(0, k^2 S_0 - K)$ . But in this model there are several market conditions for the second-stage state of the market, and in different market conditions the probability of the rising or falling of the stock price is not same, so we cannot use the simple binary tree model directly to discount the end of the second-stage of several possible option prices. Considering the possibility of the second-stage market, the value of each possibility may be different. In this paper, we use the weighted average of these possibilities as the first-stage option pricing formula:

$$\bar{C}_{1j} = \sum_{i=-1}^1 P_{0i} C_{1j}^i, \quad i = -1, 0, 1, j = 1, 2 \quad (7)$$

Where  $C_{1j}^i$  denotes the  $j$ -th possible option price under the  $i$ -th kind of market condition. Thus,  $C_{11}^i = \frac{u^i C_{21} + (1 - u^i) C_{22}}{1 + r^i}$ ,  $C_{12}^i = \frac{u^i C_{22} + (1 - u^i) C_{23}}{1 + r^i}$ , where  $i$  represents the value of the state of the market is  $-1, 0, 1$ . Thus having a zero time option price

$$\begin{aligned} C_0 &= \frac{u^0 \bar{C}_{11} + (1 - u^0) \bar{C}_{12}}{1 + r^0} \\ &= \frac{1}{1 + r_0} \left[ u^0 \sum_{i=-1}^1 P_{0i} \frac{u^i C_{21} + (1 - u^i) C_{22}}{1 + r^i} + (1 - u^0) \sum_{i=-1}^1 P_{0i} \frac{u^i C_{22} + (1 - u^i) C_{23}}{1 + r^i} \right] \\ &= \frac{1}{1 + r_0} \left[ u^0 \sum_{i=-1}^1 P_{0i} \frac{u^i C_{21}}{1 + r^i} + u^0 \sum_{i=-1}^1 P_{0i} \frac{(1 - u^i) C_{22}}{1 + r^i} + (1 - u^0) \sum_{i=-1}^1 P_{0i} \frac{u^i C_{22}}{1 + r^i} + \right. \\ &\quad \left. (1 - u^0) \sum_{i=-1}^1 P_{0i} \frac{(1 - u^i) C_{23}}{1 + r^i} \right] \\ &= \frac{u^0}{1 + r_0} \sum_{i=-1}^1 \frac{P_{0i} u^i}{1 + r^i} C_{21} + \frac{u^0}{1 + r_0} \sum_{i=-1}^1 \frac{P_{0i} (1 - u^i)}{1 + r^i} C_{22} + \frac{1 - u^0}{1 + r_0} \sum_{i=-1}^1 \frac{P_{0i} u^i}{1 + r^i} C_{22} + \\ &\quad \frac{1 - u^0}{1 + r_0} \sum_{i=-1}^1 \frac{P_{0i} (1 - u^i)}{1 + r^i} C_{23}. \end{aligned} \quad (8)$$

where  $C_{21} = \max(0, k^2 S_0 - K)$ ,  $C_{22} = \max(0, h k S_0 - K)$ ,  $C_{12} = \max(0, h^2 S_0 - K)$ .

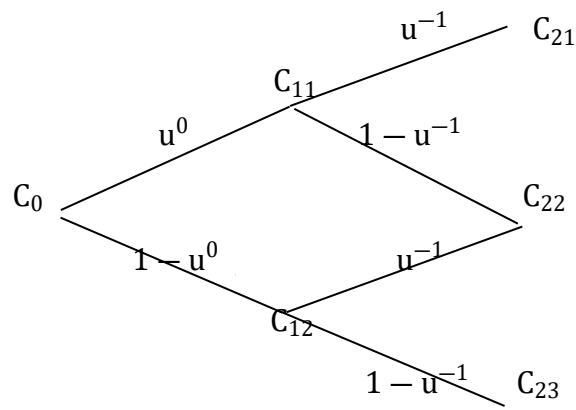


Figure 6.1 Price chart of the option when the market condition turns bad in the second period

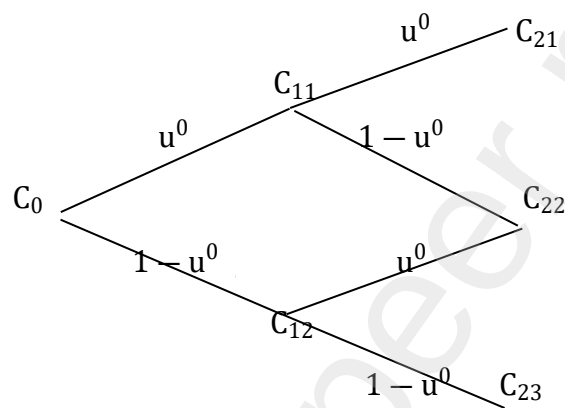


Figure 6.2 Price chart of the option when the market condition is still stable in the second period

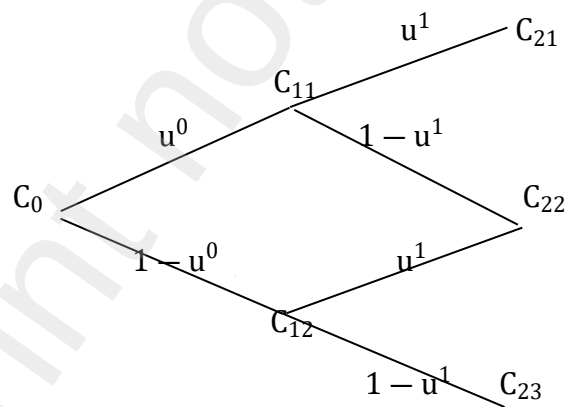


Figure 6.3 Price chart of the option when the market condition is good in the second period

This Figure 6 shows that although there may be three different scenarios while the stock or option may appear to have the same value situation at the last point in time, the probability of a stock or option value appearing is different, and this will affect the pricing of zero-time options.

### iii. The Pricing Formula of n-period European Call Option

The probabilities of values of stocks or options at the end of each period are the same, but the probabilities of those values have changed since the previous two periods, so the calculation of the probability becomes so complicated for our pricing formula. Moreover, when pricing the option at zero time, the discount formula dose not use a fixed-rate interest rate simply, but a weighted discount with different interest rates under different market conditions, the present value of option n prices should be calculated as well. As with the simple binary tree model, the possible prices of the n-terminal options are, respectively

$$C_{nj} = \max (0, k^{j-1} h^{n-j+1} S_0 - K), \quad n=1,2,\dots, j = 1, 2, \dots, n, n+1.$$

Thus, using the option pricing formula of the first period, we can express the value of the n-period option pricing formula. Let

$$\bar{U} = \sum_{i=-1}^1 \frac{P_{0i} u^i}{1+r^i}, \quad \underline{U} = \sum_{i=-1}^1 \frac{P_{0i} (1-u^i)}{1+r^i}$$

then

$$\begin{aligned} C_0 &= \frac{u^0 C_{11} + (1-u^0) C_{12}}{1+r^0} = \frac{u^0}{1+r^0} [\bar{U} \bar{C}_{21} + \underline{U} \bar{C}_{22}] + \frac{1-u^0}{1+r^0} [\bar{U} \bar{C}_{22} + \underline{U} \bar{C}_{23}] \\ &= \frac{u^0}{1+r^0} \bar{U} \bar{C}_{21} + \left( \frac{1-u^0}{1+r^0} \bar{U} + \frac{u^0}{1+r^0} \underline{U} \right) \bar{C}_{22} + \frac{1-u^0}{1+r^0} \underline{U} \bar{C}_{23} \\ &= \frac{u^0}{1+r^0} \bar{U} [\bar{U} \bar{C}_{31} + \underline{U} \bar{C}_{32}] + \left( \frac{1-u^0}{1+r^0} \bar{U} + \frac{u^0}{1+r^0} \underline{U} \right) [\bar{U} \bar{C}_{32} + \underline{U} \bar{C}_{33}] + \frac{1-u^0}{1+r^0} \underline{U} [\bar{U} \bar{C}_{33} \\ &\quad + \underline{U} \bar{C}_{34}] \} \\ &= \frac{u^0}{1+r^0} \bar{U}^2 \bar{C}_{31} + \left( \frac{1-u^0}{1+r^0} \bar{U}^2 + \frac{2u^0}{1+r^0} \bar{U} \underline{U} \right) \bar{C}_{32} + \left( \frac{1-u^0}{1+r^0} \bar{U} \underline{U} + \frac{u^0}{1+r^0} \underline{U}^2 \right) \bar{C}_{33} + \frac{1-u^0}{1+r^0} \underline{U}^2 \bar{C}_{34} \\ &\quad \dots \\ &= \frac{u^0}{1+r^0} \bar{U}^{n-1} C_{n1} + \left[ (n-1) \frac{u^0}{1+r^0} \bar{U}^{n-2} \underline{U} + \frac{1-u^0}{1+r^0} \bar{U}^{n-1} \right] C_{n2} + \dots + \\ &\quad \left( C_{n-1}^{n-j} \frac{u^0}{1+r^0} \bar{U}^{n-j} \underline{U}^{j-1} + C_{n-1}^{n-j+1} \frac{1-u^0}{1+r^0} \bar{U}^{n-j+1} \underline{U}^{j-2} \right) C_{nj} + \dots + \\ &\quad \left[ \frac{u^0}{1+r^0} \bar{U}^{n-1} + (n-1) \frac{1-u^0}{1+r^0} \bar{U} \underline{U}^{n-2} \right] C_{nn} + \frac{1-u^0}{1+r^0} \bar{U}^{n-1} C_{nn+1} \end{aligned} \quad (9)$$

Where  $\bar{C}_{gl}$ ,  $g = 1, 2, \dots, n-1, l = 1, 2, \dots, g, g+1$  is the average expected price of the first stock option in period  $g$ .  $C_{nj} = \max(0, k^{j-1} h^{n-j+1} S_0 - K)$ ,  $n=1, 2, \dots, j = 1, 2, \dots, n, n+1$  is all kinds of stock option prices at the end time point  $t_n$ .

## 5. Empirical Analysis

### 5.1 Measuring volatility

The binary option pricing formula involves the following parameters: volatility, remaining maturity  $T$ , underlying asset price  $S_0$ , strike price  $K$  and risk-free interest rate  $r$ . Among them, the volatility parameter can be used to analyze the magnitude and probability of stock's rise and fall. In order to understand these parameters, we need to model and analyze the stock returns. Based on the Markov regime-switching model proposed by Hamilton (1989), we develop a second-order, three-regimes Markov switching autoregressive model for the log rate of return  $r_t$ , hereafter referred to as MS-AR(2).

$$r_t - m_{X_t} = \phi_{1X_t}(r_{t-1} - m_{X_{t-1}}) + \phi_{2X_t}(r_{t-2} - m_{X_{t-2}}) + \varepsilon_t, \varepsilon_t | X_t \sim N(0, d_{X_t}^2),$$

$$X_t = 1, 2, 3 \quad (10)$$

In this model,  $X_t$  is the market state,  $m_{X_t}$  is the conditional expectation,  $d_{X_t}^2$  is the conditional variance of the disturbance term,  $\phi_{1X_t}$  and  $\phi_{2X_t}$  is the regression coefficient of MS-AR(2) model. Its logarithmic returns correspond to three states, expanding, contracting and smooth regimes. The corresponding parameters are different in different regime. Further, based on the regression equation (9), we can derive the expectation and variance formulas under different regime corresponding to the logarithmic return of the stock.

$$E(r_t|X_t) = m_{X_t}, Var(r_t|X_t) = \frac{d_{X_t}^2(1 - \phi_{2X_t})}{(1 - \phi_{1X_t} - \phi_{2X_t})(1 + \phi_{1X_t} - \phi_{2X_t})(1 + \phi_{2X_t})} \quad (11)$$

we can get the  $Var(r_t|X_t)$  for different regime as volatility:  $Var(r_t|X_t) = \sigma_{X_t}^2$ . Ordinary least squares methods are unable to estimate the parameters of the Markov Switching models and need to be calculated using the EM algorithm. The parameters of the model can be estimated using the MSwM package in R language.

## 5.2 Data selection and Other Parameter Settings

Basically, options in China include two major categories: real options and financial options, and financial options mainly include SSE 50 ETF options and Harvest CSI 300 ETF options. In this study, we select the data of stock returns corresponding to SSE 50 ETF and option prices as the core sample data.

In order to ensure the reliability of the results, we choose the weekly returns with a time span of more than one year for modeling and logarithmize the calculated weekly returns with the specific data time span from 2022-01-07 to 2023-08-04.

Shanghai Interbank Offered Rate (Shibor) is used as the risk-free rate. Since stock returns use weekly data, we use 7-day Shibor data that corresponds one-to-one with the date of the stock return and logarithmize it.

In order to increase the breadth of the study, a total of 19 call option contracts with different strike prices and a total of three expiration dates are selected, whose corresponding remaining expiration dates T and strike prices can be found in the website, and the stock price corresponding to the date of the initial option price  $C_0$  is selected as the initial stock price  $S_0$ . The actual option price and the stock price can be tracked from the portal of <http://xueqiu.com>. Table 1 shows some specific data.

**Table 1: Some parameter settings about the model**

| T   | K    | S0    | C0     |
|-----|------|-------|--------|
| 181 | 2.35 | 2.538 | 0.3068 |
| 181 | 2.30 | 2.538 | 0.2652 |
| ... | 2.65 | 2.538 | 0.114  |
| 90  | 2.40 | 2.538 | 0.316  |
| ... | 2.60 | 2.538 | 0.0919 |
| 55  | 2.30 | 2.538 | 0.2089 |
| ... | 2.60 | 2.538 | 0.0646 |

### 5.3 Empirical study and comparative analysis

We use the R language (see <https://CRAN.R-project.org/package=MSwM> for details) to regress the stock returns and derive the volatility, mean, and transfer probability matrix corresponding to each of the three regime. Based on the regression results, we can calculate the variance and mean corresponding to the three regimes.

$$\hat{m}_1 = \frac{c_1}{1 - \hat{\phi}_{11} - \hat{\phi}_{12}} = 0.0144, \hat{\phi}_{11} = 0.0211, \hat{\phi}_{12} = -0.0244;$$

$$\hat{m}_2 = \frac{c_2}{1 - \hat{\phi}_{21} - \hat{\phi}_{22}} = 0.0152, \hat{\phi}_{21} = -0.3787, \hat{\phi}_{22} = 0.4652;$$

$$\hat{m}_3 = \frac{c_3}{1 - \hat{\phi}_{31} - \hat{\phi}_{32}} = -0.0371, \hat{\phi}_{31} = -0.0762, \hat{\phi}_{32} = -0.1236 (c_1 \text{ and } c_2$$

are the intercepts of MS-AR(2) model,  $\hat{\phi}_{ij}$  is the regression coefficients, where  $i$  stands for the regime and  $j$  stands for the number of orders).

Next, we can calculate the expectation and volatility from these values:  $\hat{\mu}_1 = 0.0144, \hat{\sigma}_1 = 0.0105; \hat{\mu}_2 = 0.0152, \hat{\sigma}_2 = 0.0157; \hat{\mu}_3 = -0.0371, \hat{\sigma}_3 = 0.0199$ ,  $\hat{\mu}_2 > \hat{\mu}_1 > \hat{\mu}_3$  means that the regime 2 is an expanding regime, the regime 1 is a smooth

regime and the regime 3 is a contracting regime.  $\hat{\sigma}_3^2 > \hat{\sigma}_2^2 > \hat{\sigma}_1^2$  indicates that the volatility of the contracting regime is higher than the expanding regime. Also, table 3 shows the elements of transition probability matrix:  $p_{00} = 0.332747, p_{01} = 0.000012, p_{0-1} = 0.667241; p_{10} = 0.000058, p_{11} = 0.333557, p_{1-1} = 0.666384; p_{-10} = 0.684404, p_{-11} = 0.195210, p_{-1-1} = 0.120387$ .

However, the coefficients of the autoregressive model are jointly insignificant. When there is no significant autocorrelation in the series, the sample mean converges to the overall mean in the case of large samples. Therefore, we can use the sample mean and variance to measure volatility through the following equations, then,  $\hat{\mu} = -0.00265, \hat{\sigma}^2 = 0.024$  and the specific results of the regression parameters are shown in the table 2, for this table, it is clear that the coefficients of the Markov regime-switching autoregressive model are significant.

**Table 2: Parameter estimates of MS-AR model and AR model**

|                         | MS-AR model           |                       |                        |
|-------------------------|-----------------------|-----------------------|------------------------|
|                         | Regime 1              | Regime 2              | Regime 3               |
| Intercept               | 0.0112***<br>(0.0021) | 0.0221***<br>(0.0065) | -0.0244***<br>(0.0053) |
| $r_{t-1}$               | 0.3950***<br>(0.0828) | -0.3787**<br>(0.1457) | 0.4652*<br>(0.1841)    |
| $r_{t-2}$               | -0.1707<br>(0.1130)   | -0.0762<br>(0.1263)   | -0.1236<br>(0.1263)    |
| Residual standard error | 0.0097                | 0.0146                | 0.0180                 |

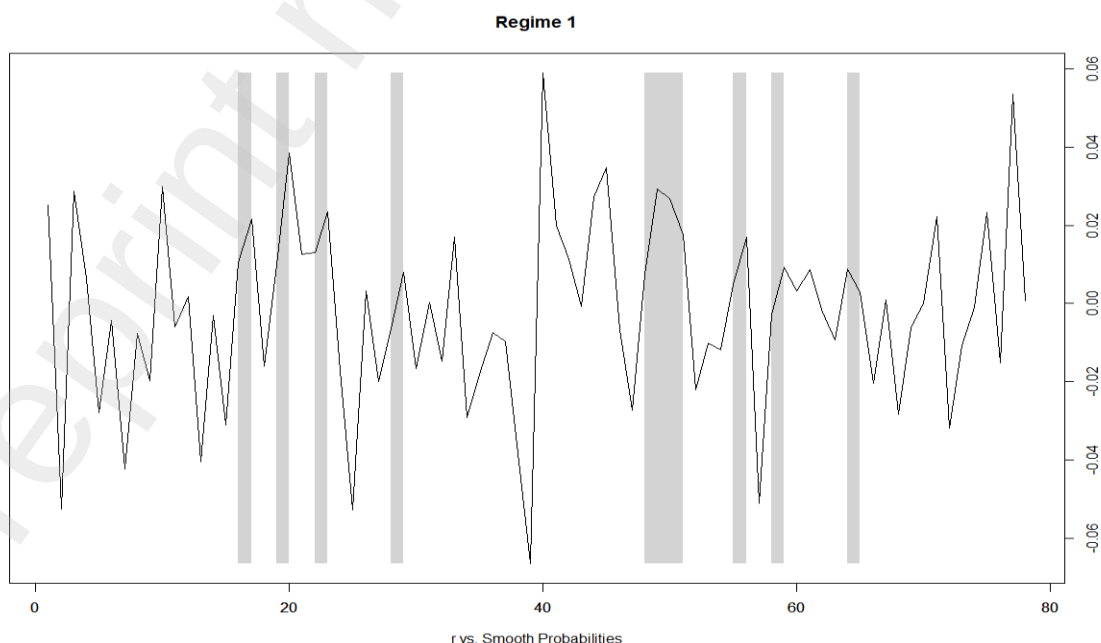
|                    |           |        |        |
|--------------------|-----------|--------|--------|
| Multiple R-squared | 0.4487    | 0.4323 | 0.2188 |
| AIC                | -361.2149 |        |        |
| BIC                | -300.7942 |        |        |
| Log Likelihood     | 189.6075  |        |        |

Note: \*\*\*, \*\*, and \* denote significant at 1 %, 5%, and 10% levels, respectively, and the figures in parentheses are standard errors.

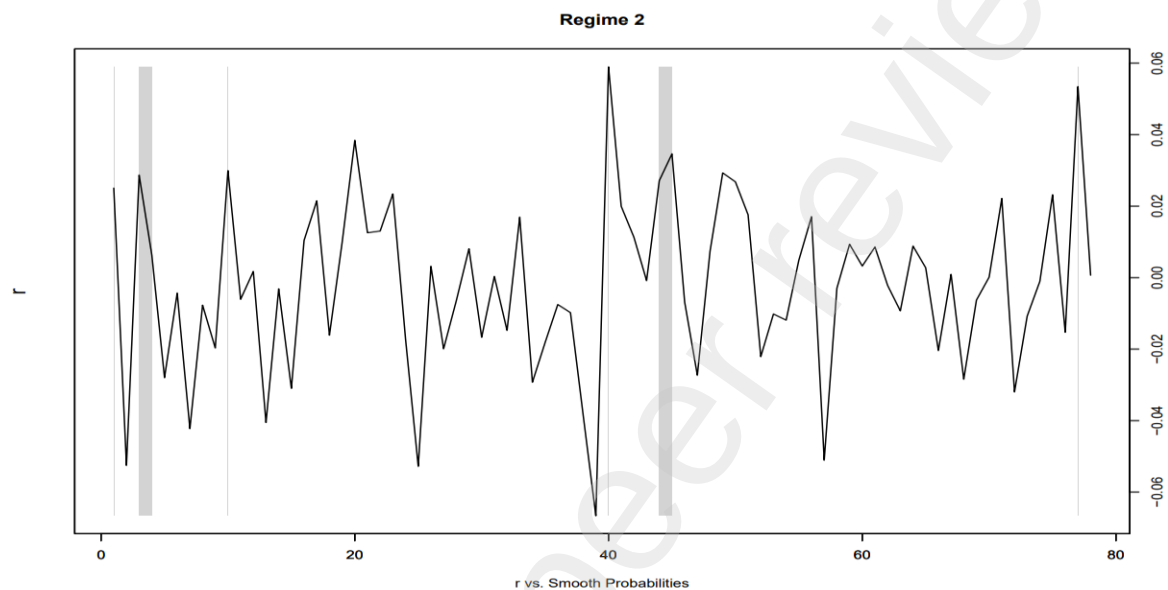
**Table 3: The transition probability matrix**

|          | Regime 1 | Regime 2 | Regime 3 |
|----------|----------|----------|----------|
| Regime 1 | 0.332747 | 0.000058 | 0.684404 |
| Regime 2 | 0.000012 | 0.333557 | 0.195210 |
| Regime 3 | 0.667241 | 0.666384 | 0.120387 |

In addition, the Markov regime-switching Model is able to distinguish which variables are related to regime 1, which are related to regime 2, and which are related to regime 3. Based on the regime-related variables shown in the figure (those gray areas in figure 7), we can calculate the value of the risk-free rate corresponding to the



different regime, in that,  $r^{(0)} = 0.0182, r^{(1)} = 0.0198, r^{(-1)} = 0.0184$  and  $r = 0.0188$ .



**Figure 7: Response variable indicating which observations are associated to regime 1&2**

After calculating the average risk-free rate and volatility, we can calculate the theoretical option prices corresponding to the specified strike prices under the regime-switching model and the classical non-regime-switching model. We also calculate the gap between the two theoretical prices and the actual price. Figure 8 shows the variation of the gap between the two theoretical prices and the actual price with respect to the strike price  $K$  for different steps  $n$ . The difference between the two theoretical prices and the actual price for different steps  $n$  is shown in the following figures. As shown in the figure 8 below, the gap between the theoretical and actual prices under the classical acyclic volatility model is larger than that under the regime volatility model. In other words, the new option pricing model based on the regime-switching framework is

closer to reality than the classic option pricing model.

In the empirical example, we argue that there are three inter-switchable market

states and the parameters vary with the market states. We not only need to distinguish

between stocks in expansion and contraction states, but also need to estimate the

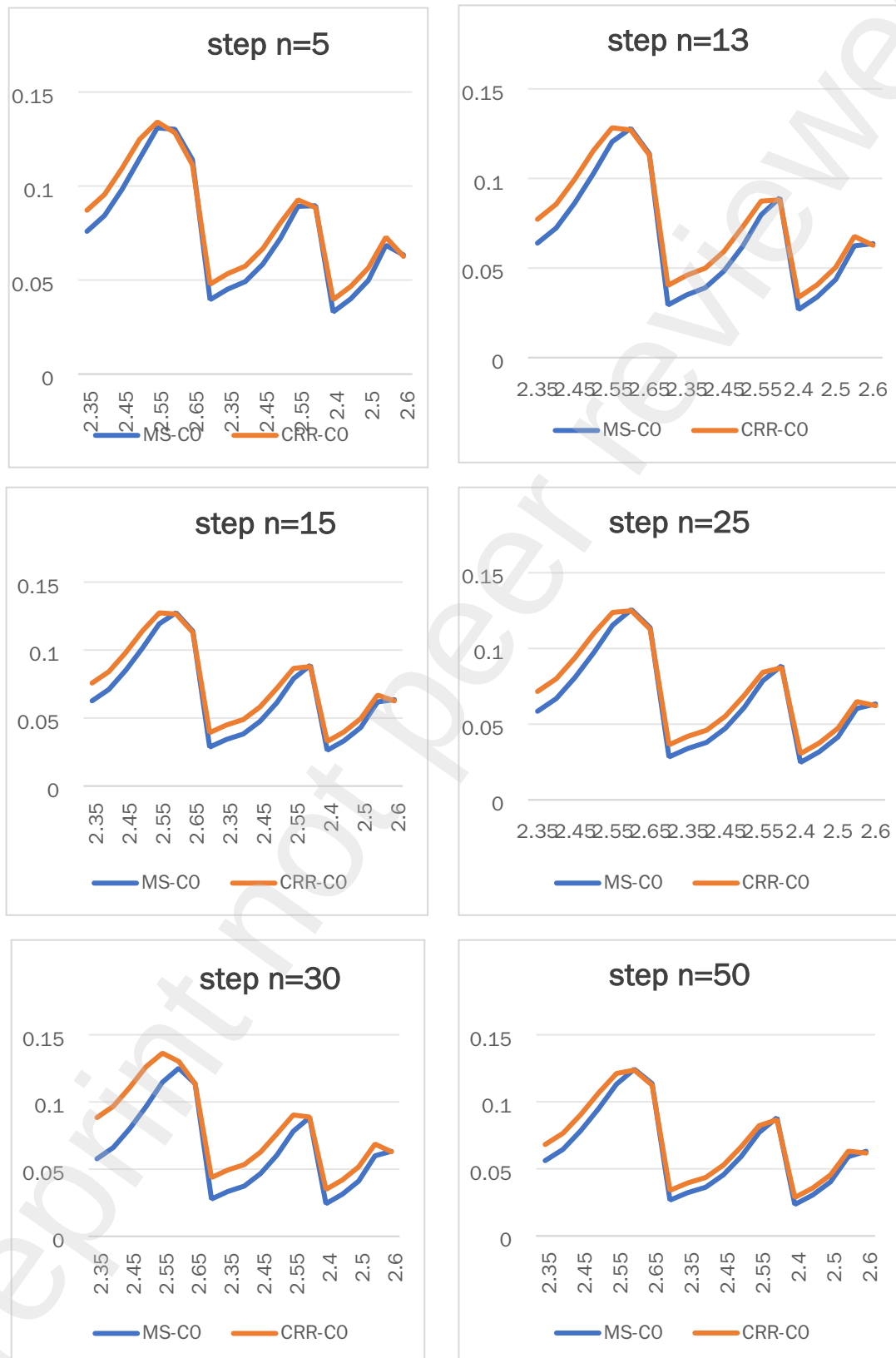
corresponding volatility and risk-free rate. Therefore, we develop the Markov regime-

switching autoregressive models to estimate these parameters. A similar approach can

be used to estimate the parameters in a classic option pricing model. We then use these

parameters to compute option prices under different strike prices for these two models.

The comparison results show that the option pricing model under regime switching



outperforms the classical option pricing model in narrowing the gap with reality.

**Figure 8: Variation of the gaps between the theoretical prices and the real prices with different strike prices  $K$  &  $T$  for different steps  $n$**

## **6. Conclusion**

It fully indicates that the regime switching pricing model is more conducive to theoretical research and market pricing analysis by several aspects from the initial theoretical assumptions of the model to the mathematical results analysis and the realistic significance of the results. Also in this paper, the model forms and calculation methods can give some reference to the finance and financial derivatives pricing which depend on the regime-switching market. We can summarize the follows:

1. The article considered that the financial market has only three different forms which are good, middle and bad, and these forms will change at any time in reality. Further, this paper considered the different market states will have different effects on the movement of the stock, in the three different states of the market, there will be different reflections and movements on the asset prices. Without doubt, the transformation of different market states is objective existence, and it affects every aspect of the economy, especially in the financial market. It is believed that the model assumption of this paper can be used for reference and applied directly to many other financial problems which are affected by the market states. And for concrete study, according to the different economic phenomena and economic environment, we also can have different numbers of the state space of Markov Models such as two, four or more.

2. In this article, after the market state was considered, the possible value of asset prices in the future period (i.e. stock or option price) could be identical. But the possibility of existence for the corresponding values will be changed. Thus, we were more inclined to use known information to estimate the future situation and then complete the current asset pricing, it is essential difference from the simple binary tree model which obtain the future stock, even the possible form of option price, and then discount the price to get the current price.

3. When this regime switching model is used, different market conditions have different transition probability matrix and motional forms of asset prices, all of which

are represented by the parameters which need to be estimated in the specific market environment. Of course, it also involves the problem about concrete model parameter estimation method when the regime switching model is applied.

4. If the option needs to be priced at a time between the start time and the end of a non-occurring event, we cannot discount the option price at expiration directly, since the pricing needs to be based on the relevant information at the time, namely the stock price. . If the state of the market or the price of the stock cannot be confirmed before time  $t$ , we may also need to further weigh on average by mathematical method.

5. In reality, the change of market state will not only affect the probability of stocks but also affect the range of the stock's rising and falling, which is not considered in this article, and that will be our future research direction.

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